Prayas JEE 2025

Practice Sheet (Mathematics)

Sequence and Series

Q1 If the sides of a right angled triangle from an AP, the
sines of the acute angles are
(A) $\frac{3}{5}, \frac{4}{5}$
(B) $\sqrt{3}, \frac{1}{3}$
(C) $\sqrt{\frac{\sqrt{5}-1}{2}}, \sqrt{\frac{\sqrt{5}+1}{2}}$
(D) $\frac{\sqrt{3}}{2}, \frac{1}{2}$

Q2 The sixth term of an AP is equal to 2 . The value of the common difference of the AP which makes the product $a_1 a_4 a_5$ least, is
(A) $8/5$
(B) $5/4$
(C) $2/3$
(D) None of these

Q3 If the arithmetic progression whose common difference is non-zero, the sum of first $3n$ terms is equal to the sum of the next n terms. The ratio of the sum of the first $2n$ terms of the next $2n$ terms is
(A) $1/5$
(B) $2/3$
(C) $3/4$
(D) None of these

Q4 Given that x, y, z are positive reals such that $xyz = 32$. The minimum value of $x^2 + 4xy + 4y^2 + 2z^2$ is equal to :
(A) 64
(B) 256
(C) 96
(D) 216

Q5 Consider the pattern shown below:

$$\begin{array}{llllll} \text{Row} & 1 & 1 & & & \\ \text{Row} & 2 & 3 & 5 & & \\ \text{Row} & 3 & 7 & 9 & 11 & \\ \text{Row} & 4 & 13 & 15 & 17, & 19, \text{ etc} \end{array}$$

The number at the end of row 60 is
(A) 3659
(B) 3519
(C) 3681
(D) 3731

Q6 Let a_n be the nth term of an AP. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, the common difference of the AP is
(A) $\alpha - \beta$
(B) $\beta - \alpha$
(C) $\frac{\alpha-\beta}{2}$
(D) None of these

Q7 If a_1, a_2, a_3, a_4, a_5 are in HP, then $a_1 a_2 + a_2 a_3 + a_3 a_4 + a_4 a_5$ is equal to
(A) $2a_1 a_5$
(B) $3a_1 a_5$
(C) $4a_1 a_5$
(D) $6a_1 a_5$

Q8 If a, b, c and d are four positive real numbers such that
$abcd = 1$, the minimum value of $(1+a)(1+b)(1+c)(1+d)$ is
(A) 1
(B) 4
(C) 16
(D) 64

Q9 If a, b, c are in AP and $(a+2b-c)(2b+c-a)(c+a-$ b) $= \lambda abc$, then λ is
(A) 1
(B) 2
(C) 4
(D) None of these

Q10 If $a_1, a_2, a_3, \ldots$ are in GP with first term a and common
ratio r, then $\frac{a_1 a_2}{a_1^2 - a_2^2} + \frac{a_2 a_3}{a_2^2 - a_3^2} + \frac{a_3 a_4}{a_3^2 - a_4^2} + \ldots + \frac{a_{n-1} a_n}{a_{n-1}^2 - a_n^2}$ is equal to
(A) $\frac{nr}{1-r^2}$
(B) $\frac{(n-1)r}{1-r^2}$
(C) $\frac{nr}{1-r}$

(D) $\frac{(n-1)r}{1-r}$

Q11 The sum of the first ten terms of an AP is four times the
sum of the first five terms, the ratio of the first term to the common difference is

(A) $1/2$ (B) 2
(C) $1/4$ (D) 4

Q12 If $\cos(x-y), \cos x$ and $\cos(x+y)$ are in HP, the
$\cos x \sec\left(\frac{y}{2}\right)$ is equal to

(A) $\pm\sqrt{2}$
(B) $\frac{1}{\sqrt{2}}$
(C) $-\frac{1}{\sqrt{2}}$
(D) None of these

Q13 If 11 AMs are inserted between 28 and 10 , the number of integral AMs is

(A) 5 (B) 6
(C) 7 (D) 8

Q14 If x, y, z are in $GP(x, y, z>1)$, then $\frac{1}{2x+\ln x}, \frac{1}{4x+\ln y}, \frac{1}{6x+\ln z}$ are in W.

(A) AP
(B) GP
(C) HP
(D) None of these

Q15 The minimum value of the quantity $\frac{(a^2+3a+1)(b^2+3b+1)(c^2+3c+1)}{abc}$ where $a, b, c \in R^{+}$, is

(A) $\frac{11^3}{2^3}$
(B) 125
(C) 25
(D) 27

Q16 Suppose, α, β are roots of $ax^2+bx+c=0$ and γ, δ are roots of $Ax^2+Bx+C=0$.
If $\alpha, \beta, \gamma, \delta$ are in AP, then common difference of AP is

(A) $\frac{1}{4}\left(\frac{b}{a}-\frac{B}{A}\right)$
(B) $\frac{1}{3}\left(\frac{b}{a}-\frac{B}{A}\right)$
(C) $\frac{1}{2}\left(\frac{c}{a}-\frac{B}{A}\right)$
(D) $\frac{1}{3}\left(\frac{c}{a}-\frac{C}{A}\right)$

Q17 Suppose, α, β are roots of $ax^2+bx+c=0$ and γ, δ
are roots of $Ax^2+Bx+C=0$.
If a, b, c are in GP as well as $\alpha, \beta, \gamma, \delta$ are in GP, then A, B, C are in

(A) AP only
(B) GP only
(C) AP and GP
(D) None of these

Q18

	Column-I	Column-II	
(A)	If $4a^2+9b^2+16c^2=2(3ab+6bc+4ca)$, where a, b, c are non-zero numbers, then a, b, c are in	(P)	AP
(B)	If $(a^2+b^2+c^2)p^2-2p(ab+bc+cn)+(n^2+b^2+c^2) \leq 0$, where a, b, c, p, n are non-zero numbers, then a, b, c are	(Q)	GP
(C)	If $a^2+9b^2+25c^2=abc\left(\frac{15}{a}+\frac{5}{b}+\frac{3}{c}\right)$, where a, b, c are non-zero numbers, then a, b, c are in	(R)	HP

(A) (A) $\rightarrow(\mathrm{P}),(\mathrm{B}) \rightarrow(\mathrm{P}),(\mathrm{C}) \rightarrow(\mathrm{R})$
(B) (A) $\rightarrow$ (R), (B) $\rightarrow(\mathrm{Q}),(\mathrm{C}) \rightarrow(\mathrm{R})$
(C) (A) $\rightarrow$ (R), (B) $\rightarrow(\mathrm{R}),(\mathrm{C}) \rightarrow(\mathrm{Q})$
(D) (A) $\rightarrow(\mathrm{R}),(\mathrm{B}) \rightarrow(\mathrm{R}),(\mathrm{C}) \rightarrow(\mathrm{P})$

Q19 Let a, b, c, d be positive real numbers with $a<b<c<d$. Given that a, b, c, d are the first four terms of an AP and a, b, d are in GP. The value of $\frac{ad}{bc}$ is $\frac{p}{q}$, where p and q are prime numbers, then the value of q is

Q20 Sum of the infinite series $\frac{1}{3}+\frac{3}{(3)(7)}+\frac{5}{(3)(7)(11)}+\frac{7}{(3)(7)(11)(15)}+\ldots$ is

(A) $1/2$
(B) $1/3$
(C) $1/6$
(D) $1/4$

Q21 $\frac{1}{1!}+\frac{1+3}{2!}x+\frac{1+3+5}{3!}x^2\ldots$

(A) $e^x(1+x)$

(B) $e^x(1-x)$

(C) xe^x

(D) $e^x(x+2)$

Q22 If $\sum_{r=1}^{n} r^4=I(n)$ then $\sum_{r=1}^{n}(2r-1)^4$ is equal to

(A) $I(2n)-I(n)$

(B) $I(2n)+16I(n)$

(C) $I(2n)-16I(n)$

(D) $I(2n)-4I(n)$

Q23 Let $S_n=\sum_{r=1}^{n} t_r=\frac{1}{6}n\left(2n^2+9n+13\right)$, then $\sum_{r=1}^{n}\sqrt{t_r}$ equals

(A) $\frac{1}{2}n(n+1)$

(B) $\frac{1}{2}n(n+3)$

(C) $(n+1)^2$

(D) n^2

Q24 The sum of the series $\frac{12}{2!}+\frac{28}{3!}+\frac{50}{4!}+\frac{78}{5!}+\ldots$. is

(A) e

(B) 3e

(C) 4e

(D) 5e

Q25 The value of $\sum_{n=3}^{\infty}\frac{1}{n^5-5n^3+4n}=$

(A) $\frac{1}{120}$

(B) $\frac{1}{96}$

(C) $\frac{1}{24}$

(D) $\frac{1}{144}$

Q26 If $S_1,S_2,S_3,\ldots\ldots.,S_p$ are the sums of first n terms of p arithmetic progressions whose first terms are $1,2,3\ldots.p$ and whose common differences are $1,3,5,7,\ldots.$, then the value of $S_1+S_2+S_3+\ldots\ldots+S_p$ is

(A) $np(np+1)/2$

(B) $np(np-1)/2$

(C) $(np-1)(np+1)/2$

(D) $n(np+1)$

Q27 Let $S_1,S_2,S_3,\ldots$ be squares such that the length of the side of S_n is equal to the length of the diagonal of S_{n+1}. Match the items in column I with those in column II, if the length of the side of S_1 is equal to 10 units.

	Column - I		Column - II
(A)	Length of the side of S_3 is	(p)	7
(B)	Length of the diagonal of S_4	(q)	5
(C)	The area of S_n is less than 1 if n is greater than	(r)	6
(D)	Sum of the areas of the squares is	(s)	200
		(t)	$\frac{10\sqrt{2}}{(\sqrt{2}-1)}$

(A) (A)$\rightarrow$ (q), (B) $\rightarrow$ (q), (C) $\rightarrow$ (p), (D) $\rightarrow$ (s)

(B) (A) $\rightarrow$ (r), (B) $\rightarrow$ (p), (C) $\rightarrow$ (s), (D) $\rightarrow$ (q)

(C) (A) $\rightarrow$ (q), (B) $\rightarrow$ (r), (C) $\rightarrow$ (s), (D) $\rightarrow$ (p)

(D) (A) $\rightarrow$ (p), (B) $\rightarrow$ (p), (C) $\rightarrow$ (r), (D) $\rightarrow$ (q)

Q28 The sequence $a_1,a_2,a_3,\ldots$ is a geometric sequence with common ratio r. The sequence $b_1,b_2,b_3,\ldots$ is also a geometric sequence. If $b_1=1,b_2=\sqrt[4]{7}-\sqrt[4]{28}+1,a_1=\sqrt[4]{28}$ and $\sum_{n=1}^{\infty}\frac{1}{a_n}=\sum_{n=1}^{\infty}b_n$, then the value of $\left(1+r^2+r^4\right)$ is

Q29 Let $a_1=\frac{1}{2},a_{k+1}=a_k^2+a_k\forall k\geq 1$ and $x_n=\frac{1}{a_1+1}+\frac{1}{a_2+1}+\ldots.+\frac{1}{a_n+1}$

Then $[x_{100}]$ is equal to (where [.] denotes the greatest integer function.)

Q30 Match Column I with Column II

Column-I		Column-II	
(A)	In an AP, the series containing 99 terms, the sum of all the odd numberd terms is 2550. The sum of all the 99 terms of AP is	(P)	5010
(B)	f is a function for which $f(1) = 1$ and $f(n) = n + f(n-1)$ for each natural number $n \geq 2$. The value of $f(100)$ is	(Q)	5049
(C)	Suppose, $f(n) = \log_2 3 . \log_3 4 . \log_4 5 \ldots \log_{(n-1)} n$, then $\sum_{k=2}^{100} f\left(2^k\right)$ equals	(R)	5050
(D)	Conscentric circles of radii 1, 2, 3.... 100 cm is drawn. The interior of the smallest circle is colured red and the annular regions are coloured altemately green and red, so that no two adjacent regions are of the same colour. The total area of green regions in sq cm is $K\pi$, then K equlas	(S)	5100

(A) $(\mathrm{A}) \rightarrow(\mathrm{P}),(\mathrm{B}) \rightarrow(\mathrm{R}),(\mathrm{C}) \rightarrow(\mathrm{Q}),(\mathrm{D}) \rightarrow(\mathrm{P})$

(B) $(\mathrm{A}) \rightarrow(\mathrm{Q}),(\mathrm{B}) \rightarrow(\mathrm{R}),(\mathrm{C}) \rightarrow(\mathrm{Q}),(\mathrm{D}) \rightarrow(\mathrm{R})$

(C) $(\mathrm{A}) \rightarrow(\mathrm{P}),(\mathrm{B}) \rightarrow(\mathrm{R}),(\mathrm{C}) \rightarrow(\mathrm{Q}),(D) \rightarrow(\mathrm{R})$

(D) $(\mathrm{A}) \rightarrow(\mathrm{R}),(\mathrm{B}) \rightarrow(\mathrm{P}),(\mathrm{C}) \rightarrow(\mathrm{Q}),(\mathrm{D}) \rightarrow(\mathrm{P})$

Answer Key

Q1	(A)	Q16	(A)
Q2	(C)	Q17	(B)
Q3	(A)	Q18	(B)
Q4	(C)	Q19	3
Q5	(A)	Q20	(A)
Q6	(D)	Q21	(A)
Q7	(C)	Q22	(C)
Q8	(C)	Q23	(B)
Q9	(C)	Q24	(D)
Q10	(B)	Q25	(B)
Q11	(A)	Q26	(A)
Q12	(A)	Q27	(A)
Q13	(A)	Q28	7
Q14	(C)	Q29	1
Q15	(B)	Q30	(C)